Part I

Small and medium-sized enterprises (SMEs), retailers and HACCP systems

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HACCP and SMEs

Problems and opportunities

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2.1 HACCP and SMEs

2.1.1 Introduction

Practical experience and a review of food safety literature indicate that success in developing, installing, monitoring and verifying a successful HACCP system is dependent on a complex mix of managerial, organisational and technical hurdles. In coping with this set of interrelating factors, even the largest food companies, equipped with significant resources of money, technical expertise and management skills, may face a difficult challenge; the small and medium sized enterprises (SMEs) may feel that the difficulties of HACCP are potentially insurmountable.

This chapter aims to discuss general problems, propose solutions and identify opportunities for SMEs in their attempt to develop systems that are technically sound, appropriate and manageable. It is based on the experience of both practitioners and researchers and uses relevant examples wherever possible.

2.1.2 Definition of an SME

There is no single, clear and widely accepted definition of an SME but they are usually classified by number of employees, turnover and profit levels. This chapter uses EU terminology to classify medium, small and micro businesses and a summary can be seen in Table 2.1. Given that there will be many differences between the smallest and largest SME (varying from the self-employed to 250 employees), discussion will be focused on those companies with *less than 50 employees*, where there is evidence of most difficulty in applying new technology/systems in general and HACCP in particular. Such

Table 2.1 Classification of SMEs

Maximum	Micro	Small	Medium
Number of employees	10	50	250
Turnover (million ECU)		7	40
Balance sheet total (million ECU)		5	27
Max. % owned by larger company		25%	25%

Note: Larger company refers to one or several enterprises not satisfying same SME criteria.
Source: European Commission Recommendation, 3 April 1996, 96/280/EU.

businesses *generally* serve local customers; have a limited share of the available market; are owned by one person, or by a small group of people; are managed by their owners who deal with all management issues usually with little other help; and are independent businesses, not parts of, or owned by, larger companies.

2.1.3 Economic significance of SMEs

SMEs contribute substantially to the production, manufacture and retail of food in both advanced and developing countries. It is argued that they are 'everywhere essential to economic development'[1] as an integral part of all market economies. It is unfortunate that the study of the role of SMEs in economic development has been retarded by lack of reliable statistics at both national and international level. However, figures available from the UK can be used to illustrate the nature of the contribution of SMEs in relation to larger companies (see Table 2.2). It is evident that a substantial proportion of food is produced, processed and sold by SMEs and the safety of their operations affects the integrity of the entire food chain. The recognition of this fact has recently led many governments to focus attention on the control of food safety in these smaller businesses.

2.1.4 HACCP implementation in SMEs

There is increasing evidence that while HACCP is widespread in large food businesses, its use is limited within SMEs. This is reflected in recent studies in

Table 2.2 An analysis of the economic significance of SMEs in the UK (1998)

	Number of businesses	Employment	Turnover
Small	99.2%	44.7%	38.0%
Medium	0.6%	11.6%	13.8%
Large	0.2%	43.7%	48.1%

Source: Department of Trade and Industry. *Small and Medium Enterprise (SME) Statistics for the United Kingdom*, 1998, London, DTI, August 1999.

the UK and Europe which have found that SMEs are less likely to invest in hygiene and food safety than large companies and are less likely to have HACCP in place.[2,3] Indeed, one study identified that for companies with less than 50 staff, HACCP implementation decreased proportionally as number of employees decreased.[4]

This is not surprising given that the move toward HACCP implementation in many companies has been customer driven. This is very evident in UK companies that supply large retailers (e.g. supermarkets) whose contracts often require documented evidence of a HACCP system from their suppliers. This is outside the remit of most SMEs, many of whose customers are the end user (the consumer) as in the case of retailers. The only pressure to apply HACCP for these companies has been from legislation. It is well known that the risk of prosecution for food hygiene offences (within the regulatory system of most countries) is low and therefore changes in legislation provide little motivation for owners to embark on a completely new system of managing food safety. This is especially true if owner-managers do not believe they have a problem anyway – unfortunately a commonly held belief, despite a world-wide escalation of foodborne disease. This lack of motivation has been a significant reason that there has been little uptake of HACCP within small and micro businesses in particular. Other reasons, in particular resource constraints, will be discussed later in the chapter.

It must be remembered, however, that the consumer demands safe food whether it is purchased from a small, medium or large company. The application of HACCP systems (combined with GHPs) within SMEs is a major step forward in protecting public health.

2.1.5 A blueprint for SMEs?

For most SMEs, even if they could be motivated to change, their progress would be severely hindered by a lack of information as to how to apply HACCP principles and what the completed system should look like in the context of their own businesses. While there is widespread agreement that the Codex methodology* is applicable to all food operations, there is little evidence of its successful application within SMEs, particularly the small business. Indeed, attempts to make HACCP fit such businesses often involve a crude scaling down of the principles without evidence to ensure that safety has not been compromised.[5] There is an urgent need to provide high quality, authenticated models that will guide SMEs through the entire HACCP process.

In an attempt to produce such ‘blueprints’ many generic models have been developed by governments (particularly in North America) and trade organisations. However, most do not extend beyond identification of CCPs and offer no help with implementation and management of the system. There is

* The term ‘Codex methodology’ will be used to describe the application of HACCP principles following the steps outlined by the Codex Alimentarius Commission.[6]



Fig. 2.1 Retail meat managers enjoying HACCP training.

also no evidence of their validity, suitability or uptake by SMEs. Indeed, Codex have recently published specific guidelines for the application of HACCP in small businesses.[7]

A recent initiative to develop HACCP models for SMEs is the *Accelerated HACCP Project* within the UK meat industry. This is a £9 million government-funded project which aims to implement HACCP in 7,000 independent retail butchers. It was instigated as a direct response to an outbreak of *E. coli* 0157 food poisoning in Scotland, which resulted in 21 deaths. Initial evaluation of the project suggests that the strong commitment of owner-managers to safe food can be harnessed effectively if a blueprint is available and the necessary support given (see Fig. 2.1).

2.2 Setting up the HACCP system

2.2.1 Access to HACCP training

Successful HACCP implementation requires (1) competence in the application of HACCP principles and (2) technical expertise in food microbiology and food chemistry. Although these two areas overlap, they are considered separately so as to focus on the needs of the typical SME.

It is doubtful whether any company can implement HACCP without specific training. This is particularly true for the SME with limited access to information and often without the time or skills to interpret textbook scenarios. Indeed, good HACCP literature is restricted to a handful of user-friendly books amidst a plethora of watered-down manuals of limited use. As discussed earlier, the

fundamentals of HACCP methodology are as necessary for the SME as any other company, and the typical short course (two days) is an effective introduction to the concept and the jargon. However, for the SME without in-house technical support, it is important not to 'abandon trainees to their fate after the initial familiarisation is completed'.[8] Further specialist help is required which will consider the development, implementation and management of the system within the constraints of the SME.

Statistics from the UK's major provider of HACCP courses illustrate the limited uptake and availability of even introductory level training, e.g. the Royal Institute of Public Health & Hygiene's Certificate in HACCP Principles, established in 1995 – currently 31 centres in the UK with 4,220 candidates to date. Given that there are 600,000 food premises within the UK, the majority of these being SMEs, this equates to a severe skills shortage. Evidence of this lack of skill has been confirmed not only within food businesses but among the consultants and enforcement officers to whom many SMEs turn for help with implementation.

The expense and efficacy of consultants was investigated in a recent research project within UK SMEs.[9] Companies were paying between £160 and £1,200 for varying degrees of input and several companies said that consultancy had been poor value for money and would not be used in future. More significantly, when the HACCP system in each business was independently audited,* none of the plans that had been developed with consultants were considered adequate.

2.2.2 Technical expertise

The typical SME, as discussed in section 2.1, is often described as having limited resources of staff, time, skills, expertise and finance and therefore has to be dependent on external support. In the context of HACCP this is particularly relevant in terms of technical expertise. Whereas competency in HACCP methodology can be effectively gained through training, this must be complemented with the appropriate knowledge of food microbiology and food chemistry. While the Codex Decision Tree and other such hazard analysis tools are readily available, they rely on the underlying technical expertise of the user.

In SMEs, particularly the micro business, technical competency to undertake hazard analysis is rarely available in-house. A recent government-funded case study of 24 SMEs involved in manufacturing ready-to-eat meat products identified that the employment of experienced technically qualified personnel was the single most important factor influencing the implementation of HACCP.[9] Analysis of company size indicated that this was not the norm for companies with less than 25 staff or a turnover of less than £2 million – in effect, most SMEs. Indeed, research consistently shows that the majority of SME managers have received no more than a one-day basic introduction to food

* The audits were conducted by the European Food Safety Inspection Service (EFSIS) accredited auditors from the Campden & Chorleywood Food Research Association.

hygiene (e.g. the Chartered Institute of Environmental Health's Basic Food Hygiene Qualification). Some of the consequences of such a skills shortage include the following:

- A lack of ability to prioritise the risks from physical, microbiological and chemical hazards. Given the time involved in hazard analysis, it is important for SMEs to focus HACCP studies on the group(s) of hazards that pose the greatest threat to public health. For example, caterers will focus their efforts on microbiological hazards while the soft-fruit grower will be more concerned with foreign body contamination. Many SMEs are unable to make these decisions and attempt to study all groups of hazards at once: a process that invariably ends in confusion, overload and a dilution of control. This scenario impinges on the wider debate as to how risk is perceived by, and communicated to, both the food industry and the consumer.
- The inability to distinguish between the relative risks of different pathogens on particular foods. This leads directly to the common scenario where the HACCP studies involve every pathogen 'in the textbook'. For example, the sandwich manufacturer who spent many weeks developing a HACCP plan to control the growth of *Clostridium botulinum* on lettuce leaves – an anaerobe that would never grow in the presence of air.
- A lack of focus at the stage of hazard identification is compounded by the inability to make technical decisions as to criticality. This results in too many CCPs being identified. This is a problem in many large organisations also, but its root within these companies is usually the misinterpretation of methodology. This can easily be remedied with HACCP training. For the SME the problem is insoluble without recognition of, and access to, technical expertise.
- Given the nature of control needed at a CCP, it is evident that the system can only operate with small numbers of CCPs. Hence the identification of too many CCPs results in a dilution of control and, in effect, a systems failure. An example best illustrates this point. During a recent training session a technical manager from an SME revealed that on a single process line he had identified 143 CCPs. Further investigation revealed that the Company HACCP Manual had been followed explicitly: every process step had to have at least one hazard and every hazard was critical! The reason that he had attended the training was that the system was impossible to implement. A subsequent review established only four CCPs and control was established within weeks.
- Interpretation of microbiological and chemical data on such documents as Certificates of Analysis is another problem area for the SME. Without a thorough understanding of both the science and statistics such documents can easily mislead. For example, a snack manufacturer was buying garlic flavouring on the basis that each batch came accompanied with a Certificate of Analysis to show the absence of *Salmonella* in 25 g of product. Only after a major food poisoning outbreak, identifying this as the source, did the sampling regime come into question: one 25 g sample was tested per tonne of product!

Another factor, perhaps more important than lack of technical knowledge, is that it is often not even recognised. In the UK it is quite normal for someone with production experience to attend a short course on basic food hygiene and subsequently feel competent to undertake a HACCP study involving high risk foods. This over-confidence can be a danger, especially when practices have continued for many years, and is reflected in the saying so often heard from small business owners, 'I've been doing this for 30 years and I've never killed anyone yet.'

2.2.3 Team approach

The SME invariably adapts well to teamwork, bypassing many problems found in larger companies. The relatively small number of employees enables the 'depth' and 'breadth' of the organisation to be represented within one team, enabling greater ownership to develop. It is also likely that managers have worked their way up from the factory floor (and continue to work there when production demands are high) and this first-hand knowledge of operational procedures and empathy for the workforce speeds up the process and accuracy of CCP identification. The subsequent control strategies are likely to be both practical and effective.

For example, a small factory producing garlic butter had a problem of fragments of blue plastic lining from the butter packs getting into the final product. The HACCP team included five people with production experience. While the focus was initially on how to detect the plastic, the storeman quickly identified the cause of the problem as one of staff fatigue when moving large, heavy blocks of butter encased in wood. In the relaxed environment of a 'production' rather than 'management' led team, he was able to admit that the problem arose when blocks were dropped, wood shattered and fragments tore into the plastic with the result that plastic fragments became embedded in the butter. The storeman suggested a very effective control which was subsequently implemented – job rotation so that the amount of heavy lifting was restricted to short time periods. While this scenario may fit any organisation with a good HACCP team, in reality many large companies still write HACCP plans in head office, thus missing the vital role of operatives.

The smaller the SME, the more difficult it is to release staff to attend meetings. It may be necessary to reduce the team size to a bare minimum and coopt members as and when required. For the micro business this still does not solve the problem because even the core team may be equivalent to more than half of the total workforce. In such circumstances innovative methods of teamwork are needed.

For example, a small hospital catering unit employed a total of eight staff. The catering manager was fully committed to HACCP, given the vulnerable group in her care, but could not persuade the hospital managers to pay the staff for half an hour extra per week for team meetings. She therefore resorted to 'clipboard meetings'. While supervising production she would go to each

member of staff in turn with her clipboard and identify, by consensus, CCPs and subsequent controls. Although she complained of sore feet and a sore throat, the HACCP system was implemented successfully in six months.

2.2.4 Flow diagrams

The development of detailed flow diagrams is an essential part of the HACCP process, whatever the size of operation. However, there is much resistance to this detail across all sectors of the food industry where a minimalist approach is usually adopted. This is especially evident for SMEs where generic models have been suggested by many national governments as a mechanism for managing the complexities of HACCP. This advice inherently undermines the identification of CCPs as it is usually at the process steps that are not depicted on simplified flow charts that potential problems are most likely to occur. For example, the linear flow depicted in many generic flow diagrams takes no account of re-work!

Another problem is that one generic process step, for example the delivery of raw materials, cannot allow for the assessment of risk of individual foodstuffs. For example, a catering organisation was following a generic flow chart which identified the delivery step as critical. The control measures they adopted involved a temperature check. This led to the time-consuming practice of measuring and documenting the incoming temperature of all perishable foods including oranges, lemons and lettuces! Again, the loss of focus seriously undermined the concept of HACCP, increased paperwork and de-motivated staff who were well aware of the futility of the task they were required to undertake.

For SMEs overwhelmed by the task of producing multiple flow charts it is important to realise that many products have common processes. By working on one HACCP flow diagram at a time these ‘overlapping areas’ become evident and ways of merging the numerous flow charts – without losing essential detail – become obvious. Thus, one flow chart may eventually represent multiple products but this does not compromise HACCP principles. Most SMEs can find one member of staff with artistic ability, purchase some paper and coloured pens and produce excellent flow diagrams if they are convinced of the necessity.

Even in the catering industry, where one outlet may have dozens of menu items, this procedure can be followed as long as the essence of HACCP is not lost, that is, focus. The identification and control of high risk items must be a priority for HACCP studies, particularly in this sector of the industry. A consultant recently related the problem of a hospital caterer who had been told by the local enforcement officer that a HACCP flow diagram and plan was needed for every menu item including yoghurt and cheese. Both these products were purchased as individual, sealed packs and were only opened on the ward, by the patients, immediately prior to consumption. The caterer was totally de-motivated as his own common sense told him this was unnecessary.

2.2.5 Relationship between good hygiene practice and HACCP

Within a food operation there are many general activities that are in place to control food safety, for example pest control, cleaning and disinfection procedures, equipment maintenance, induction training and personal hygiene practices. This general level of control, applicable across all areas of the food operation, can be considered under the generic term good hygiene practice (GHP). This has been defined as the 'basic hygiene measures which establishments should meet … for the production of food of acceptable [safety]'.[10] The terms good manufacturing practice (GMP), pre-requisite programmes (PRPs) or GHP are often used interchangeably in this context.

HACCP, in contrast, is directed at individual product/process lines and examines whether there are any additional, specific, significant hazards – above and beyond GHP – which need to be controlled to ensure product safety. The two systems are therefore inextricably linked but distinctly different. This relationship is represented diagrammatically in Fig. 2.2. An example would be the risk of contamination of *Staphylococcus aureus* during sandwich preparation in a busy catering outlet. It would be general good practice to have all food handlers trained in hygienic practice and wearing clean protective clothing. However, the staff preparing the sandwiches may take the additional precaution of wearing disposable gloves. The former measures would form part of GHP while the latter would be a specific control implemented for sandwich preparation in the HACCP plan.

In effect, GMP screens out the general hazards allowing the HACCP plan to focus on the significant hazards that need additional, specific control measure(s) to ensure product safety. Further analysis, using the decision tree or other hazard analysis tools, will determine which of these are CCPs. This screening process results in the identification of a relatively small number of CCPs – one essential criterion for successful HACCP implementation.

A recent survey[11] of 50 SMEs established that none had an effective GHP system in place. This resulted in every process step identified in the HACCP study being cluttered with repetitive, general hazards leading to excessive documentation and, more importantly, the identification of too many CCPs. Post-survey training, which fully explored the role of GHP within HACCP development, resulted in the decision of all of these companies to review and refine GHP, prior to any further work on HACCP.

It is not surprising that GHP and HACCP have become confused given the lack of written guidance on how the two systems interrelate. One exception is in North America where GMP has been a legal requirement in some sectors of the food industry since the 1970s. Interestingly enough, HACCP has yet to become a legal requirement across the breadth of the US food industry. With SMEs in mind, particularly micro businesses, the US Food and Drug Administration* has produced a detailed guide as to how food retailers can develop a system of GHP

* Details of this guide along with extensive information relating to US legislation, enforcement and practice can be accessed using the FDA website.[12]

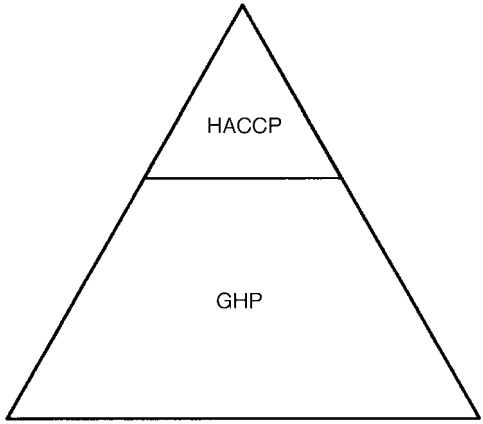


Fig. 2.2 The relationship between HACCP and GHP.

and then build on this to establish additional controls using HACCP principles. This offers a clear route to a comprehensive food safety management system and is recommended reading for all SMEs working in small premises with a large number of overlapping operations.

2.2.6 Resource problems

The typical SME can be described as having a busy day-to-day existence without designated staff to get involved in long-term planning of non-essential activities, i.e. those not directly related to production. In larger companies the training and technical departments often lead the HACCP project: most SMEs do not have these resources. It is evident therefore that even if the SMEs can be convinced of the necessity for HACCP, the allocation of sufficient time for its development becomes a major constraining factor. This is compounded by the requirement for specific HACCP training and the need to access the necessary technical expertise, as discussed in sections 2.2.1 and 2.2.2. To the SME this translates into a heavy financial burden and most owners look to government or other agencies for external help at minimal cost. While the ethics of this are debatable, the reality is that small and micro businesses, in particular, do not feel that *they* should pay for change initiated externally. Some initiatives and opportunities for SMEs are outlined below:

- The UK government, acting through an industry body, has recently attempted to support the large-scale implementation of HACCP in 7,000 retail butchers (see section 2.1.5). The project involved a two-day training course for groups of 20 butchers using material specifically adapted to suit the size and nature of the business. This was followed by eight hours of one-to-one consultancy within the butcher's own premises. While the training and consultancy were delivered free of charge, many butchers complained that 'time was money'

and it was 'costing too much'. Indeed, many had to close their business to attend the training and meet with the consultants in the evenings or on Sundays.

- A monthly food safety drop-in resource centre, catering specifically for SMEs, has been set up by a university in the north-west of England. This has been funded by a variety of government and non-government organisations. The initiative is proving highly successful for businesses as they come across specific areas of difficulty such as flow charting, identification of hazards and validation.
- Trade associations also have a part to play in the dissemination of HACCP expertise although the costs of joining have been found to act as a barrier to micro businesses. Various research organisations can also offer significant resources to food businesses but the annual fees are prohibitive to all but the largest SMEs.
- The use of the Internet is developing at an exponential rate and can offer the SME a cost-effective resource in terms of expertise and information. The industry-specific discussion groups allow SMEs to share problems and expertise and there are vast amounts of technical data on-line. This is likely to be of considerable use to SMEs in the future, particularly for those isolated in rural areas.

2.3 Operating the HACCP system

2.3.1 Documentation

One of the criticisms made by SMEs trying to operate the HACCP system is its requirement for documentation. However, researchers and practitioners confirm that the excessive documentation reported is usually associated with the system being developed inappropriately.[13] This highlights the need for effective training in HACCP methodology.

Invariably companies that employ technical staff, with dedicated time for administrative work, have found documentation requirements easier to achieve than those who rely on hands-on managers. For many such SMEs, especially micro businesses, paperwork of any kind is a burden with verbal communication playing a major role in the successful management of their businesses. However, practice has shown that even within the smallest business, HACCP record keeping can be integrated into practice (with minimal disruption) if managers believe it to make good business sense.

The message that must be sold to SMEs is that HACCP aims to ensure food safety with the *minimum* necessary control. It is a system that, if correctly applied, *focuses* control at the CCPs. If the HACCP study has correctly identified a small number of CCPs then this need not be onerous. Fig. 2.3 illustrates the key elements that will demonstrate that the CCP is under control. For most businesses, achieving this level of documentation will produce a significant improvement in food safety and a slow, developmental approach

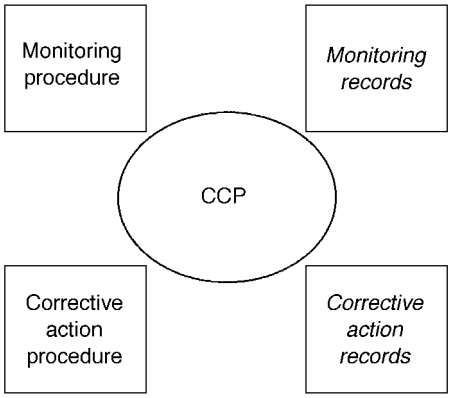


Fig. 2.3 The essential documentation required for each CCP.

should be encouraged to meet further requirements for verification and review (see below).

The use of computer software has been suggested as a good way for SMEs to deal with the development and documentation of a HACCP programme. While surveys have shown that companies with computer hardware enthusiastically purchase various packages, they subsequently find them to be of limited use.[11] Recently, various software packages specifically designed for SMEs have appeared on the market, including some for micro businesses, but it is too early to assess their usefulness. It would seem that in the long term, software may well be an appropriate solution to document control for SMEs. At the present time, however, owner-managers express the need for simple, highly visible documents such as wall charts and white boards (with pencils and pens attached) to encourage routine completion.

2.3.2 Validation

Identification of CCPs must be followed by decisions as to how they can be controlled effectively. In many companies, large and small, such decisions are often based on custom and practice rather than on evidence. HACCP should be seen as an opportunity to justify these practices using whatever means are available. SMEs need not be daunted as many CCPs are based on parameters, such as temperature, time and pH, which can be validated using simple experiments. Indeed, many SMEs enjoy this aspect of HACCP, feeling that they are *taking control* of food safety rather than being pushed into change by external forces.

For example, a self-employed butcher, while undertaking HACCP training, was frustrated by the conflicting advice he received as to safe procedures for cooling cooked hams. His trade organisation, enforcement officer, tutor and textbook all gave differing views. After the training he bought a temperature probe and plotted the temperature of a cooked ham at hourly intervals until it

reached a safe temperature. He was shocked to find that, even following the most stringent procedure recommended, this took over 17 hours. He subsequently reviewed and revised all his cooking and cooling procedures until he had confidence that his practices were safe and the subsequent monitoring was valid. This butcher, who had received no formal education for over 30 years, so enjoyed this research exercise that he enrolled at his local university on a part-time MSc in Food Safety Management.

A specific cleaning procedure is another commonly identified measure to control a CCP and the cleaning methods chosen must be shown to work. The use of Agar Immersion Plating and Contact (AIPC or 'Dip') slides have proved useful to validate the effectiveness of cleaning routines. They are cheap, readily available and, in conjunction with the manufacturer's advice, can identify effective cleaning practice.[8]

Some critical limits, for example the shelf-life of new products, will need to be validated by technical experts and the SME must access this expertise as and when necessary. Local educational institutions, trade organisations and enforcement authorities are a useful starting point. It is important for the small and micro business in particular, with limited funds, to focus on specific problem areas when seeking advice.

The monitoring of critical limits should ideally be carried out 100% of the time but this is not possible for many control measures. In practice limits are checked at predetermined intervals. Yet again, *ad hoc* decisions usually determine this frequency and HACCP teams must be encouraged to test out these decisions. For example, a baker cooking pork pies in batches of 100 always tested one pie from the right-hand side of the bottom shelf to ensure that it had reached a safe temperature. His experience told him this was necessary due to the uneven cooking temperature of the oven. After undertaking HACCP training, the baker decided to test his practice by temperature checking the entire batch on one occasion. This confirmed that the pies at the right-hand side of the bottom shelf cooked more slowly than the rest of the batch. If one of these pies had reached the desired temperature then he could safely assume that the rest of the batch had. While he had always known his practice was safe, he now had the evidence to prove it. Again, taking control of food safety in this way proved to be an empowering experience for this owner-manager.

2.3.3 Verification

Once a HACCP plan has been developed and introduced into a food operation it must be maintained on a continuous basis. It is vitally important that those involved in food production follow the requirements of the HACCP plan and record essential detail. An audit trail then exists for verification, at any time, by supervisors, managers, customers or enforcement officers.

As discussed already, one of the criticisms of HACCP made by SMEs is that it requires a great deal of paperwork with records having to be kept at all stages of the production process. Those trying to audit food operations in turn complain

of records being incomplete or distributed around the operation, making the audit more of an arduous and painstaking affair than it need be. The advantages of having a clearly thought out and carefully operated verification system are clear: it allows for audits to be quickly and accurately completed; it provides managers with all the data to hand so they can immediately act if problems occur; and it allows operatives to see the whole food system rather than just a small part and so helps them gain ownership of the HACCP system.

One method of managing verification that allows paperwork to be limited while maintaining control is by using a *build-card*, which accompanies the product as it moves through the processing operation. The build-card contains all the necessary details about the product (like a list of ingredients, delivery requirements, customer details). As the product moves along the production line all the details of its processing, such as cooking times and storage temperatures, are recorded on the build-card. The product thus ends up ready for dispatch with a complete history attached and ready for inspection by anyone with a need to know. A copy may also be sent to the customer for their records. This system has been used successfully in a cook-chill operation in England. With over 100 menu items, this was seen to be the most cost-effective solution to document control. The build-card also contained quality details and a diagram of how the meal components should be presented.

More advanced systems use bar-coding technology in order to record necessary information as the product moves along the line, and all the information is kept on a computer so that it can be accessed whenever needed. This type of bar-coding is used in an SME in Canada that produces fats and shortenings for caterers. The system allows for rigorous monitoring of all batches so that product only becomes available for release to the customer when all critical limits are within range and operating procedures have been followed correctly. Any batch, for example, that has not had a microbiological test is not registered on the warehouse system and can therefore not be released for delivery, since any product taken off the shelf by the storeman has to have its bar code scanned and the computer would instantly identify this as problematic.

In addition to routine auditing the HACCP system requires a periodic *review* to demonstrate that it is meeting its objective – producing safe food. Large companies will often employ external experts for this purpose as it is difficult to achieve the independence required using internal staff. This may well be supplemented by microbiological analysis of the food, process and environment. The costs of such activities are prohibitive to most SMEs, particularly small and micro businesses. For these companies the routine visits from enforcement officers and other agencies can be used effectively as an independent review of the safety of the system (see section 2.3.5).

Fierce protectiveness over HACCP plans, which is understandable given the labour that has gone into them, seems to prohibit small businesses from collaborating at this level. However, a network of businesses could work together and audit each other's systems with little or no investment. It is interesting that the butchers taking part in the HACCP initiative, despite being

local competitors, were seen to swap ideas and practices and even invite visits from others on the scheme. While these owner-managers are by no means technical experts, they represent a fresh pair of eyes and have much to contribute to food safety reviews. The benefits of HACCP training, by bringing together people with similar problems, cannot be overestimated.

2.3.4 'Ownership' and motivation

Technical barriers have long been quoted as the major hurdle to successful HACCP implementation. However, experience is proving that the major challenge is the development of a system that can actually work in practice, which almost invariably requires a change in the culture of the organisation.

For example, it could be argued that the most important people in the organisation (in terms of food safety) are the operatives who control the CCPs. These employees are often the least paid, least valued and least motivated – a recipe for disaster! HACCP has identified a way forward in terms of *involvement* and subsequent *ownership*. If operatives are (1) told that they are in charge of a critical process, (2) asked to join a team to develop a strategy for dealing with this, and then (3) helped to write realistic procedures in their own 'language', they may become motivated to carry out safe procedures at all times. Such *participation* in technological change and *delegation* of control to those closest to the production process are well-documented driving mechanisms in the management of change and essential to successful HACCP implementation.

It is interesting that the larger the organisation, the more difficult it is to initiate and maintain such culture change. SMEs with their less formal management structures and simpler communication channels are at a positive advantage in this respect. The smaller the organisation, the more likely it is that all those involved in HACCP will have hands-on experience. This further improves the ability of the team to develop a system that operatives and managers alike will own with subsequent commitment and motivation to make it work.

2.3.5 Third-party audits and SMEs

Without in-house technical expertise many SMEs are vulnerable to take advice from everyone and anyone. This can result in HACCP being a patchwork of CCPs developed reactively on the insistence of external auditors or enforcement officers.

For example, in a family-owned meat cutting plant a competent quality manager, after implementing HACCP, recounted a problem with a customer audit. The retailer involved was insistent that the three microbiological CCPs identified in one particular process could not possibly represent a thorough HACCP study. The bottom line was 'the next time I visit, you must have more CCPs or we will terminate your contract'. This is not an uncommon scenario and questions the competence of many third-party auditors.

It is important for companies to have confidence in their systems and challenge the legitimacy of advice from those who have not been involved with the HACCP development. At the very least it is worth asking to see a full curriculum vitae of external auditors to assess their level of competency. While some countries, for example Australia, have stringent requirements for commercial HACCP auditors (including substantial experience of the type of food operation under question), this is by no means commonplace. On the other hand, if the auditor has the relevant knowledge and experience the visit could be used to full advantage and seen as a review of the system in place. Indeed, the audit can become a cost-effective management tool rather than a threat.

2.3.6 Supplier safety assurance

HACCP can only be completely effective if, at every critical control point, real-time preventative control is maintained. While a company can do much to achieve this within its own production process, it is questionable whether there are any real-time controls that can be implemented to assure contamination-free foodstuffs bought in from a third party, particularly if it involves overseas suppliers.

Larger organisations invest considerable time and money in attempts to assure the safety of such food supplies. This often involves detailed specifications which give information as to the presence or absence of microbiological, chemical and physical hazards in the food as well as Certificates of Analysis as evidence to support these specifications and on-site audits by technical staff. They may also restrict purchasing to suppliers who have attained third-party accreditation of their HACCP system.

For the SME there are difficulties in operating any effective system of supplier vetting, given the lack of time and expertise available. The smaller the company, the greater are the problems, with many micro businesses relying on negotiation through telephone contact and often buying from middlemen who are themselves SMEs with little formal control over food safety.

Training in HACCP has been shown to have a positive effect with respect to purchasing practice. If the owner-manager can be encouraged to focus on the need for evidence of a *safe* supplier, this is a good step forward. Many HACCP plans refer to the use of 'only reputable suppliers', with little thought as to what this actually means. An investigation[14] of the purchasing criteria used by 300 professionally qualified catering managers indicated that the term 'reputable' actually related to quality, cost and locality of suppliers. Safety was not even a formal consideration for 96% of these managers. However, after training, many of them asked to see evidence of supplier HACCP plans and some visited their local operations to discuss safety. This must be seen as a significant step forward in the overall context of food safety at the SME–supplier interface.

Looking to the future, formal accreditation for all but the smallest businesses may become commonplace (if costs can be contained). This could considerably improve the process of supplier safety assurance for all SMEs. It is also likely

that many more businesses will make use of the Internet and the considerable resource it offers. At the present time model specifications for a wide variety of foodstuffs are available on the Web, giving an invaluable insight into the kind of data required from suppliers to give purchasers confidence of product safety.

2.4 Conclusions

The typical SME manager, who is often the owner, is undoubtedly highly motivated to achieve the best possible standard for the business. This commitment must be channelled into the application of HACCP principles in order to secure safe food. Despite a lack of interest and slow uptake to date, there are many encouraging developments which should help chart the way forward for SMEs.

In general terms, the nature of the risks inherent in the food industry must be communicated more effectively to both producer and consumer. This falls within the remit of local and national government and is high on the agenda in many countries. At the present time a combination of ignorance and optimism combine to block efforts to give safety the high profile necessary to stimulate change.

Current research to develop workable blueprints and benchmark best practice is encouraging, but it is important that the relatively small investments are not wasted with duplication of effort. Governments should work at an international level to coordinate activities and disseminate results. It must also be recognised that HACCP cannot be devised by government committees and needs the active cooperation and piloting among food operators themselves. It is also important to plan and deliver thorough evaluation, against predetermined success criteria. Practitioners should only be encouraged to change when there is evidence that such change is beneficial.

At a more practical level, high quality HACCP training must be widely available, at an appropriate cost, to all SMEs. While the principles remain the same for all food operations, their application needs skilled interpretation for SMEs, particularly for small and micro businesses. A developmental approach is suggested which concentrates on (1) the installation of a fully operational GHP system, (2) HACCP studies to identify specific areas that need additional control, (3) the development of valid CCP control measures and monitoring routines, and (4) appropriate systems of verification and review. This would need to be phased in, over perhaps a period of two years, and therefore relies on an ongoing support network at local level.

Each local area could develop a ‘HACCP resource centre’ which would provide some of the essential requirements for successful implementation. This may include directories of suitably qualified HACCP consultants, trainers and educational opportunities, funding routes (so often not accessed by SMEs), discussion groups to share experiences, on-hand experts to examine specific problems, and computers with access to the Internet and HACCP software.

Local government offices or educational establishments would be suitable venues. Given the dearth of technical expertise, within small and micro businesses in particular, there is an urgent need to allow access to this type of service. Perhaps SMEs themselves should lobby for such facilities within their locality.

Finally, it should hearten the SME manager to know that all companies, large and small, have problems with HACCP implementation – they are just different ones! The small company adapts well to teamwork and develops ownership with less difficulty, and their flexibility allows practical solutions to be implemented quickly and often at very little cost. If appropriate technical expertise can be accessed and good HACCP training undertaken, they should be very enthusiastic about the task ahead.

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